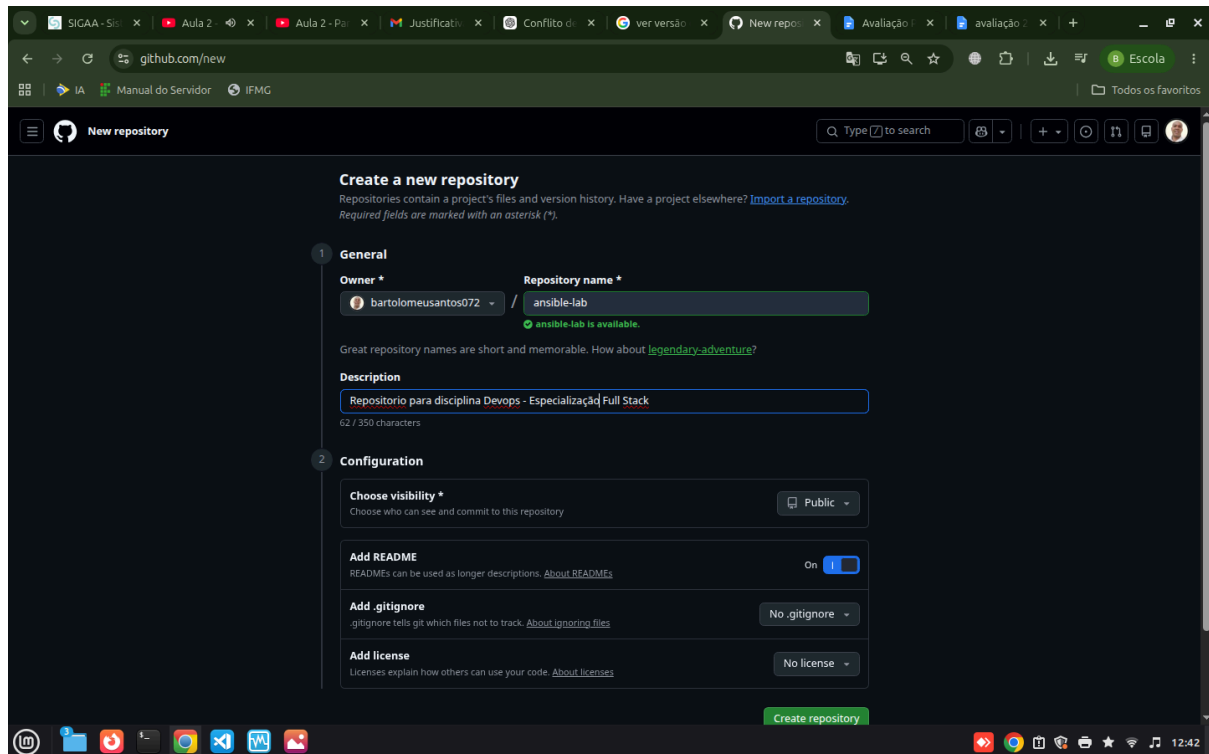
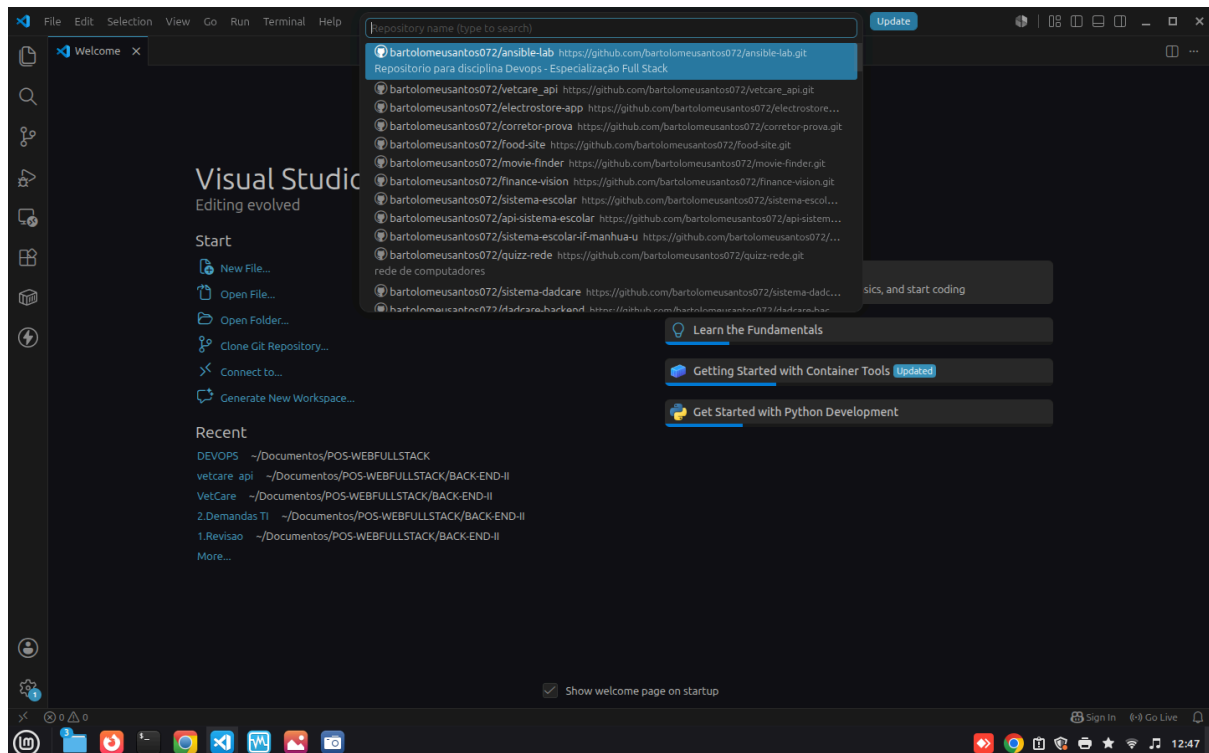


Prints de tudo

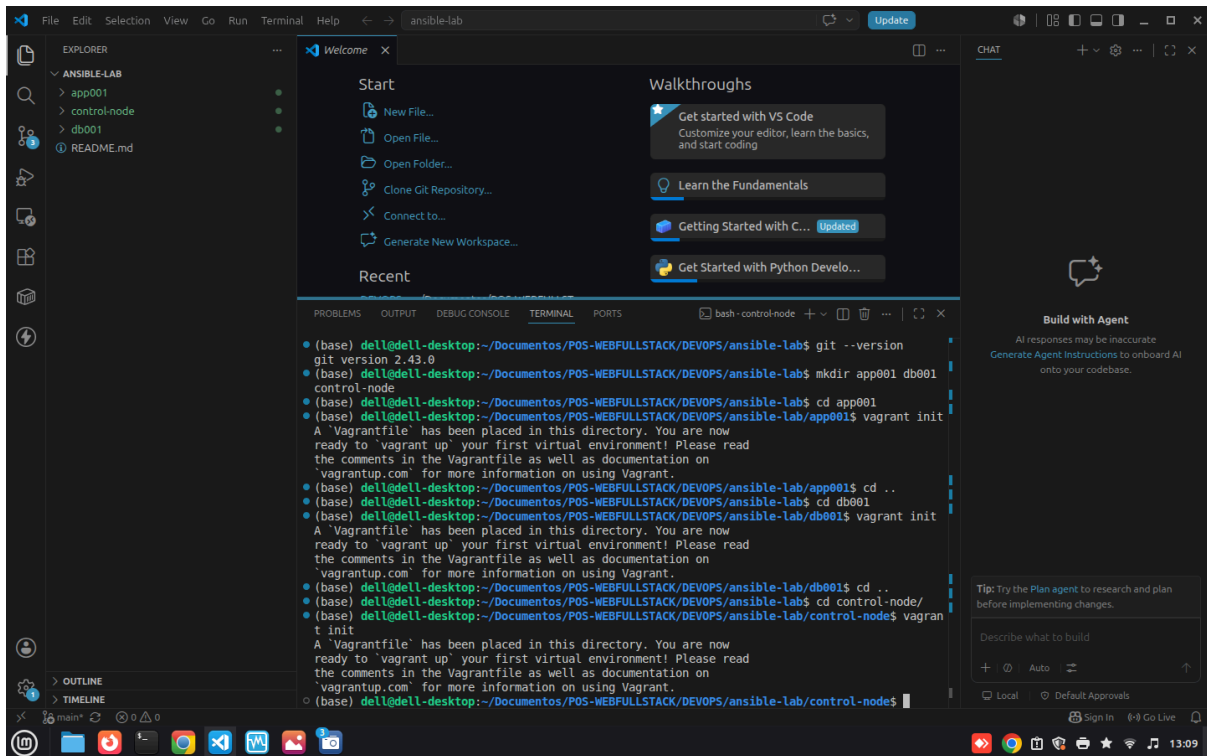
1. Criar o repositório no github



2. clonar repositório



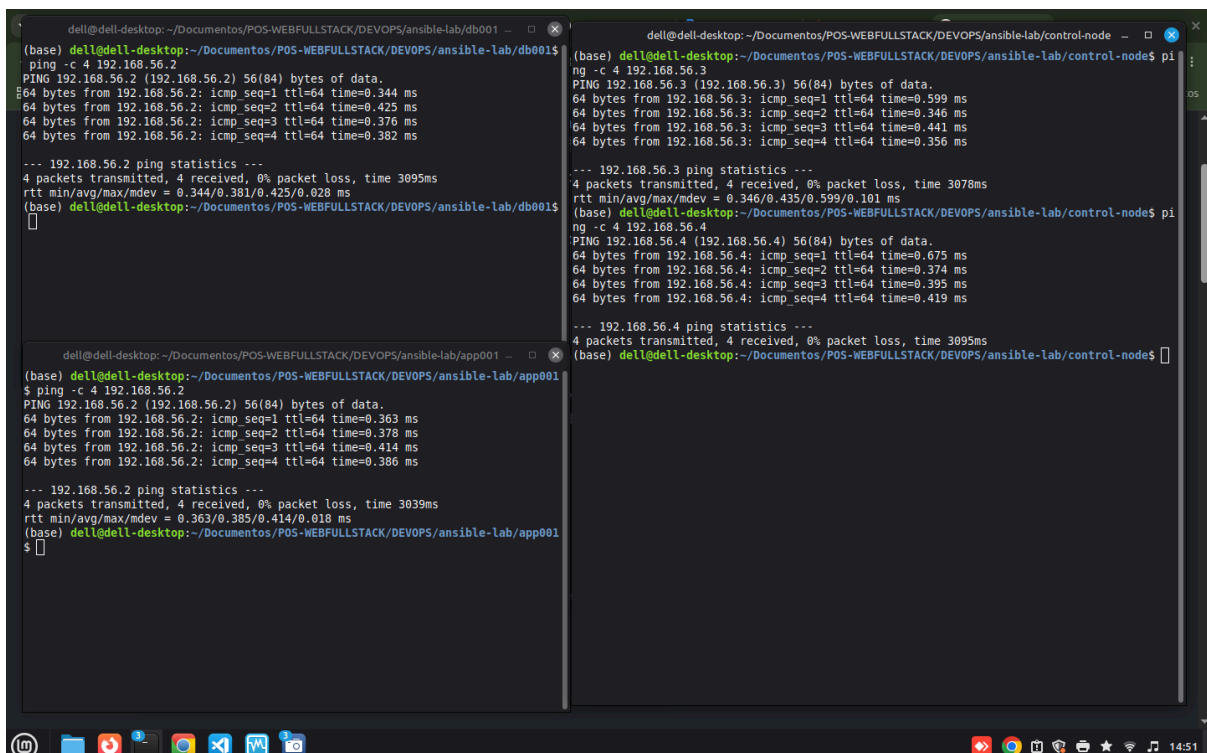
3. criar pastas db001 app001 control-node e iniciar vagrant init



The screenshot shows the Visual Studio Code interface with a workspace named 'ansible-lab'. The Explorer panel on the left shows the folder structure: 'ansible-lab' containing 'app001', 'control-node', and 'db001'. The main editor area shows the terminal output of the following commands:

```
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab$ git --version
git version 2.43.0
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab$ mkdir app001 db001 control-node
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab$ cd app001
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$ vagrant init
A 'Vagrantfile' has been placed in this directory. You are now ready to 'vagrant up' your first virtual environment! Please read the comments in the Vagrantfile as well as documentation on 'vagrantup.com' for more information on using Vagrant.
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$ cd ..
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab$ cd db001
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/db001$ vagrant init
A 'Vagrantfile' has been placed in this directory. You are now ready to 'vagrant up' your first virtual environment! Please read the comments in the Vagrantfile as well as documentation on 'vagrantup.com' for more information on using Vagrant.
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/db001$ cd ..
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab$ cd control-node/
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/control-node$ vagrant init
A 'Vagrantfile' has been placed in this directory. You are now ready to 'vagrant up' your first virtual environment! Please read the comments in the Vagrantfile as well as documentation on 'vagrantup.com' for more information on using Vagrant.
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/control-node$
```

4. teste de ping nas maquinas app001, db001, control-node



The screenshot shows two terminal windows. The left window shows the results of a ping test from the host to the 'db001' virtual machine, and the right window shows the results of a ping test from the host to the 'app001' virtual machine.

Left Terminal (db001):

```
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/db001$ ping -c 4 192.168.56.2
PING 192.168.56.2 (192.168.56.2) 56(84) bytes of data.
64 bytes from 192.168.56.2: icmp_seq=1 ttl=64 time=0.344 ms
64 bytes from 192.168.56.2: icmp_seq=2 ttl=64 time=0.425 ms
64 bytes from 192.168.56.2: icmp_seq=3 ttl=64 time=0.376 ms
64 bytes from 192.168.56.2: icmp_seq=4 ttl=64 time=0.382 ms

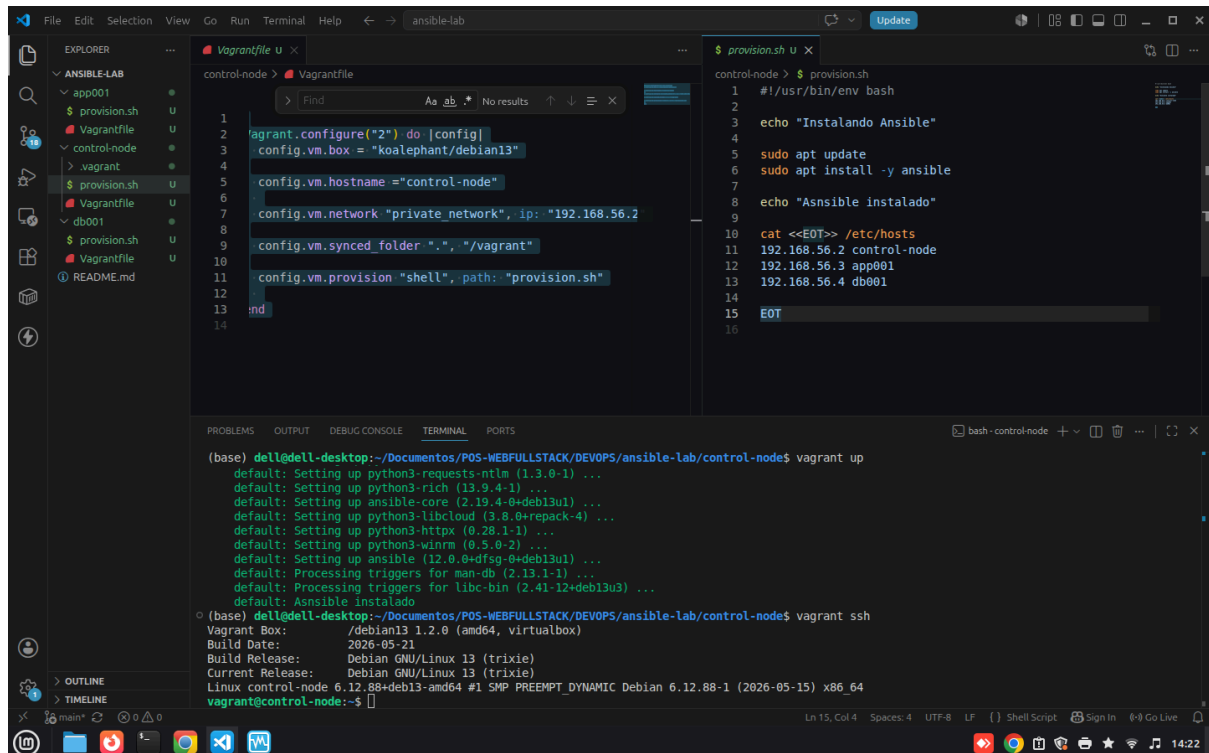
--- 192.168.56.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3095ms
rtt min/avg/max/mdev = 0.344/0.381/0.425/0.028 ms
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/db001$
```

Right Terminal (app001):

```
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$ ping -c 4 192.168.56.2
PING 192.168.56.2 (192.168.56.2) 56(84) bytes of data.
64 bytes from 192.168.56.2: icmp_seq=1 ttl=64 time=0.363 ms
64 bytes from 192.168.56.2: icmp_seq=2 ttl=64 time=0.378 ms
64 bytes from 192.168.56.2: icmp_seq=3 ttl=64 time=0.414 ms
64 bytes from 192.168.56.2: icmp_seq=4 ttl=64 time=0.386 ms

--- 192.168.56.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3039ms
rtt min/avg/max/mdev = 0.363/0.385/0.414/0.018 ms
(base) dell@dell-desktop:~/Documents/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$
```

Criar, editar o [provision.sh](#) e Vagrantfile do control-node, executar o `vagrant up`, `vagrant ssh`.

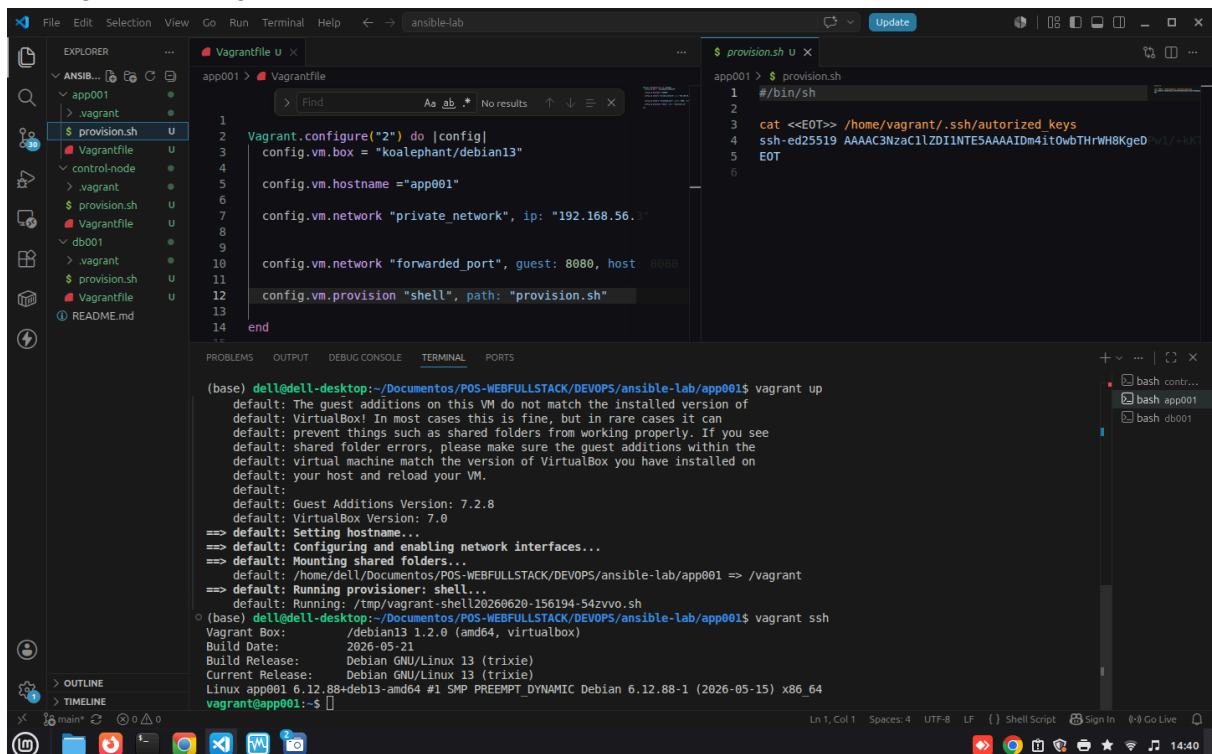


```
1 #!usr/bin/env bash
2
3 echo "Instalando Ansible"
4
5 sudo apt update
6 sudo apt install -y ansible
7
8 echo "Ansible instalado"
9
10 cat <<EOT>> /etc/hosts
11 192.168.56.2 control-node
12 192.168.56.3 app001
13 192.168.56.4 db001
14
15 EOT
```

```
(base) dell@delldesktop:~/Documentos/POS-WEBFULLSTACK/DEVOPS/ansible-lab/control-node$ vagrant up
default: Setting up python3-requests-ntlm (1.3.0-1) ...
default: Setting up python3-rich (13.9.4-1) ...
default: Setting up ansible-core (2.19.4-0+deb13u1) ...
default: Setting up python3-libcloud (3.8.0+repack-4) ...
default: Setting up python3-httpx (0.28.1-1) ...
default: Setting up python3-winrm (0.5.0-2) ...
default: Setting up ansible (12.0.0+dfsg-0+deb13u1) ...
default: Processing triggers for man-db (2.13.1-1) ...
default: Processing triggers for libc-bin (2.41-12+deb13u3) ...
default: Ansible instalado
(base) dell@delldesktop:~/Documentos/POS-WEBFULLSTACK/DEVOPS/ansible-lab/control-node$ vagrant ssh
Vagrant Box: /debian13 1.2.0 (amd64, virtualbox)
Build Date: 2026-05-21
Build Release: Debian GNU/Linux 13 (trixie)
Current Release: Debian GNU/Linux 13 (trixie)
Linux control-node 6.12.88+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.88-1 (2026-05-15) x86_64
vagrant@control-node:~$
```

Esta versão de virtualbox bloqueia a criação de redes *Host-Only* fora da faixa padrão (que é 192.168.56.0/21). Como o arquivo Vagrantfile tenta forçar o IP 192.168.1.2, o VirtualBox barra a inicialização.

5. executar `vagrant init`, editar Vagrantfile, criar editar [provision.sh](#) colar a chave publica. executar `vagrant up`, `vagrant ssh` da pasta app001 e db001



```
1 #/bin/sh
2
3 cat <<EOT>> /home/vagrant/.ssh/authorized_keys
4 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDm41t0wbThrWH8KgeD...
5 EOT
```

```
(base) dell@delldesktop:~/Documentos/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$ vagrant up
default: The guest additions on this VM do not match the installed version of
default: VirtualBox! In most cases this is fine, but in rare cases it can
default: prevent things such as shared folders from working properly. If you see
default: shared folder errors, please make sure the guest additions within the
default: virtual machine match the version of VirtualBox you have installed on
default: your host and reload your VM.
default:
default: Guest Additions Version: 7.2.8
default: VirtualBox Version: 7.0
==> default: Setting hostname...
==> default: Configuring and enabling network interfaces...
==> default: Mounting shared folders...
default: /home/dell/Documentos/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001 => /vagrant
==> default: Running provisioner: shell...
default: Running: /tmp/vagrant-shell20260620-156194-54zvo.sh
(base) dell@delldesktop:~/Documentos/POS-WEBFULLSTACK/DEVOPS/ansible-lab/app001$ vagrant ssh
Vagrant Box: /debian13 1.2.0 (amd64, virtualbox)
Build Date: 2026-05-21
Build Release: Debian GNU/Linux 13 (trixie)
Current Release: Debian GNU/Linux 13 (trixie)
Linux app001 6.12.88+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.88-1 (2026-05-15) x86_64
vagrant@app001:~$
```

7. aplicar o ansible -m ping all

8. playbooks

The screenshot shows the Ansible Lab interface with three playbooks open in the editor:

- db.yml**:

```
1 ---
2 - hosts: dbs
3   become: yes
4
5 tasks:
6
7   - name: Atualizar repositórios
8     apt:
9       update_cache: yes
10
11  - name: Instalar MariaDB
12    apt:
13      name:
14        - mariadb-server
15      state: present
16
17  - name: Iniciar MariaDB
18    apt:
19      name:
20        - mariadb-server
21      state: started
22      enabled: yes
23
24  - name: Criar bando de dados
25    shell: |
```
- app.yml**:

```
1 ---
2 - hosts: apps
3   become: yes
4
5 tasks:
6
7   - name: Atualizar repositórios
8     apt:
9       update_cache: yes
10
11  - name: Instalar Apache e PHP
12    apt:
13      name:
14        - apache2
15        - php
16        - php-mysql
17      state: present
18
19  - name: Iniciar Apache
20    service:
21      name: apache2
22      state: started
23      enabled: yes
24
25  - name: Copiar Index
26    copy:
27      src: ../plataforma/index.html
28      dest: /var/www/html/index.html
29
30  - name: Copiar processa.php
31    copy:
32      src: ../plataforma/processa.php
33      dest: /var/www/html/processa.php
34
35  - name: Copiar styles.css
```
- main.yml**:

```
1 ---
2 - import_playbook: db.yml
3 - import_playbook: app.yml
```

The terminal at the bottom shows the command `vagrant@control-node:/$`.

9. platafoma

The screenshot shows the Ansible Lab interface with three files open in the editor:

- processa.php**:

```
1 <?php
2
3 $host = "192.168.1.4";
4 $user = "cadastro";
5 $password = "123456";
6 $database = "cadastro";
7
8 $conn = new mysqli($host, $user, $password, $database);
9
10 if ($conn->connect_error) {
11     die("Erro de conexão: " . $conn->connect_error);
12 }
13
14 $nome = $_POST["nome"];
15 $email = $_POST["email"];
16 $telefone = $_POST["telefone"];
17
18 $sql = "INSERT INTO pessoas(nome, email, telefone) VALUES('$nome', '$email', '$telefone')";
19
20 $result = $conn->query($sql);
21
22 if ($result === TRUE) {
23     echo "Cadastro realizado com sucesso!";
24 } else {
25     echo "Erro ao cadastrar: " . $conn->error;
26 }
27
28 $conn->close();
```
- index.html**:

```
1 <!DOCTYPE html>
2 <html lang="pt-BR">
3 <head>
4   <meta charset="UTF-8">
5   <title>Cadastro de Pessoas</title>
6   <link rel="stylesheet" href="style.css">
7 </head>
8 <body>
9
10 <div class="container">
11
12   <h2>Cadastro de Pessoas</h2>
13
14   <form action="processa.php" method="POST">
15
16     <input
17       type="text"
18       name="nome"
19       placeholder="Nome"
20       required>
21
22     <input
23       type="email"
24       name="email"
25       placeholder="E-mail"
26       required>
27
28     <input
29       type="text"
30       name="telefone"
31       placeholder="Telefone"
32       required>
33
34     <button type="submit">
35       Cadastrar
36     </button>
37   </form>
38 </div>
39 </body>
40 </html>
```
- style.css**:

```
1 body {
2   font-family: Arial, sans-serif;
3   background-color: #f5f5f5;
4 }
5
6 .container {
7   width: 400px;
8   margin: 50px auto;
9   background-color: white;
10  padding: 25px;
11  border-radius: 10px;
12  box-shadow: 0 0 10px #000;
13 }
14
15 h2 {
16   text-align: center;
17   color: #333;
18 }
19
20 input {
21   width: 100%;
22   padding: 10px;
23   margin-top: 10px;
24   margin-bottom: 15px;
25   box-sizing: border-box;
26 }
27
28 button {
29   width: 100%;
30   padding: 12px;
31   background-color: #007bff;
32   color: white;
33   border: none;
34   border-radius: 5px;
35   cursor: pointer;
```

The terminal at the bottom shows the command `vagrant@control-node:/$`.

10. Executar playbooks

```
vagrant@control-node: /vagrant/playbooks$ ansible-playbook -i /etc/ansible/hosts main.yml

PLAY [dbs] *****

TASK [Gathering Facts] *****
[WARNING]: Host 'db001' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [db001]

TASK [Atualizar repositórios] *****
ok: [db001]

TASK [Instalar MariaDB] *****
ok: [db001]

TASK [Iniciar MariaDB] *****
ok: [db001]

TASK [Criar bando de dados] *****
changed: [db001]

TASK [Criar usuario cadastro] *****
changed: [db001]

TASK [Permitir conexões remotas] *****
changed: [db001]

TASK [Reiniciar MariaDB] *****
changed: [db001]

TASK [Copiar script SQL] *****
changed: [db001]

TASK [Criar tabela pessoas] *****
changed: [db001]

PLAY [apps] *****

TASK [Gathering Facts] *****
[WARNING]: Host 'app001' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [app001]

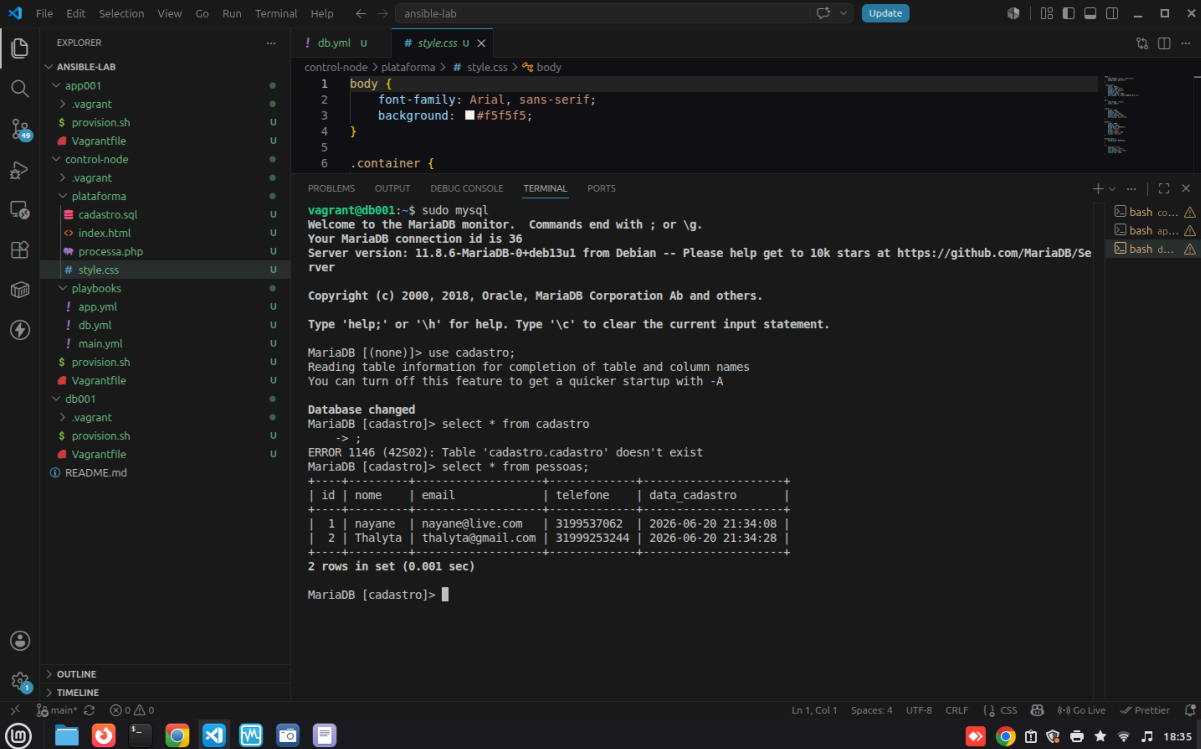
TASK [Atualizar repositórios] *****
changed: [app001]

TASK [Instalar Apache e PHP] *****
changed: [app001]
```

11. cadastrar dados

The screenshot shows a web browser window with the address bar displaying '192.168.56.3/index.html'. The page contains a form titled 'Cadastro de Pessoas'. The form has three input fields: 'Nome', 'E-mail', and 'Telefone'. A blue button labeled 'Cadastrar' is at the bottom of the form. A tooltip with the text 'Preencha este campo.' is visible over the 'Nome' field. The browser's address bar shows 'Não seguro' and '192.168.56.3/index.html'. The browser's tab bar shows several open tabs, including 'SIGAA', 'ansible', 'SIGAA', 'Aula 3', 'Aula 2', 'Justifi', 'Conflit', 'bartol', 'avalia', 'IPs de', 'Cadastr', and 'Escola'. The browser's status bar shows 'Todos os favoritos'.

12. Testar banco de dados



The screenshot shows a Visual Studio Code editor with the Ansible Lab extension. The Explorer pane on the left shows the project structure, including files like `db.yml`, `style.css`, `provision.sh`, `processa.php`, `cadastro.sql`, `index.html`, `playbooks`, `app.yml`, `main.yml`, `db001`, and `README.md`. The main editor area shows the `style.css` file with the following content:

```
1 body {
2   font-family: Arial, sans-serif;
3   background: #f5f5f5;
4 }
5
6 .container {
```

The terminal pane at the bottom shows a session with the MariaDB command-line client. The user has executed the following commands:

```
vagrant@db001:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 36
Server version: 11.8.6-MariaDB-0+deb13u1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use cadastro;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [cadastro]> select * from cadastro
-> ;
ERROR 1146 (42502): Table 'cadastro.cadastro' doesn't exist
MariaDB [cadastro]> select * from pessoas;
+----+-----+-----+-----+-----+
| id | nome  | email          | telefone | data cadastro |
+----+-----+-----+-----+-----+
| 1  | nayane | nayane@live.com | 3199537062 | 2026-06-20 21:34:08 |
| 2  | Thalyta | thalyta@gmail.com | 31999253244 | 2026-06-20 21:34:28 |
+----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [cadastro]>
```

The terminal output shows the error message "ERROR 1146 (42502): Table 'cadastro.cadastro' doesn't exist" and the successful execution of the `select * from pessoas;` query, which returns two rows of data.

<https://github.com/bartolomeusantos072/ansible-lab>